hierarchical structure, prior, parameterization, transformation

(tbc)
discrete and continuous
time-series seasonality

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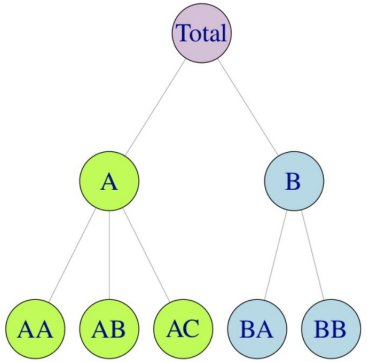
Hierarchy as axiom

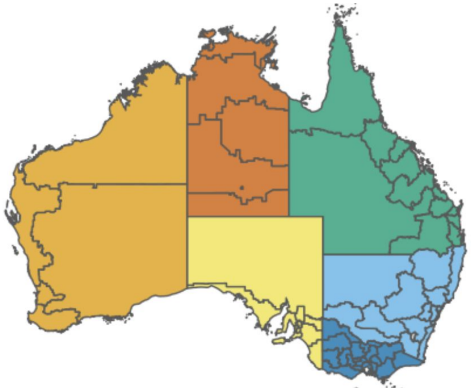
All generative systems can be expressed as a hierarchical structure.

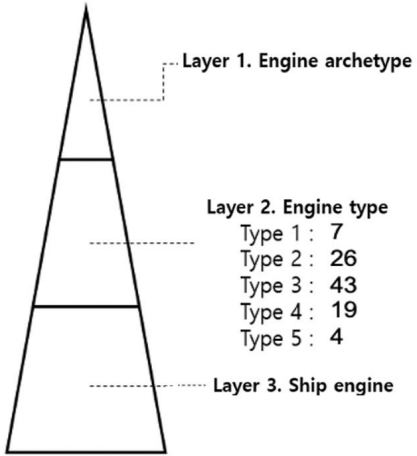
Starting from one node (layer 1), imagine a tree with three layers that branch into two child nodes each.

"Hierarchical structure" as our axiom enforces "ordering of closeness" among terminal nodes. Two nodes with a shared parent in a higher layer would have a higher correlation.



Hierarchy structure





e.g. Australia tourist demand, Ship engine failure
hierarchy structured by region, shared engine

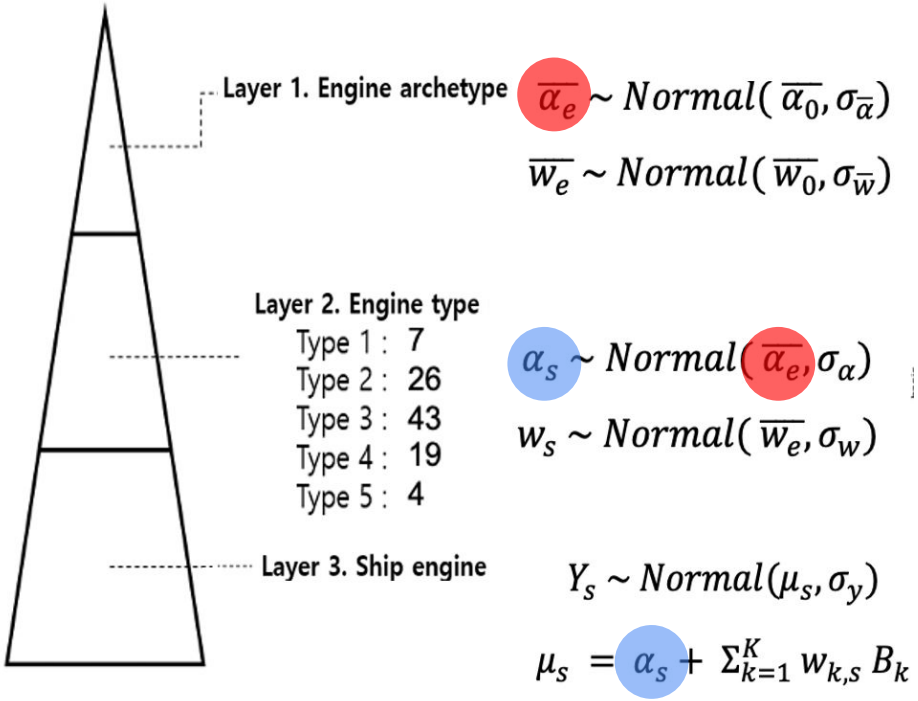
Prior at each node

Prior is **heterogeneity switch**; "H" is lenient to a heterogeneous child node while "L" forbids it.

Another name of this role is "**shrinkage**".

"L" is useful for sparse data when possible categories (e.g. day X product X store of a demand time series) are much greater than the data. First adopted "L" prior may gradually evolve to "H" with data inflow.

Prevalent custom of fixed parameter value e.g. **θ = 1 is the tightest prior**: point mass distribution. This forbids diverging and creates a single child node in our hierarchy structure.



Effect of linear transformation with parameterized prior

When linear transformations are applied to predictors in a classical regression or generalized linear model, it is for reasons of **computational efficiency** or **interpretive convenience**.

Two familiar examples are the expression of **indicators** in an ANOVA setting in terms of orthogonal contrasts and **standardization of continuous predictors to have mean 0 and variance 1**. These and other linear transformations can make regression coefficients easier to understand, but they have no effect on the predictive inferences from the model. From an inferential standpoint, **linear transformations do nothing in classical regression**, which is one reason why statistical theory has focused on nonlinear transformations. In classical inference, reparameterizations such as centering affect the likelihood for the model parameters but do not affect predictive inference.

In contrast, even simple linear transformations can substantively change inferences in hierarchical Bayesian regressions. For a simple example, consider a regression predicting students' grades from a set of pretests, in which the coefficients \beta_j for the pretests j are given a common N(\mu, \tau^2) prior distribution, where **\mu and \tau are hyperparameters estimated from the data**. Further suppose that the pretests were originally on several different scales and that each pretest has been rescaled to the range 0–100.

If the tests measure similar abilities, then it would make sense for the coefficients for the rescaled test scores to be similar. The **Bayesian model will then perform more shrinkage on the new scale and lead to more accurate predictions compared with the untransformed model**. This is an example of a reparameterization that has no effect on the likelihood but is potentially important in Bayesian inference.

more in [Parameterization in Bayesian Modeling] by Andrew (one of his fav. paper)

How prior explains the effect of FFT preprocessing for NN

FFT = neural network layer with no activation function and no bias.

i.e. FFT preprocessing is one layer of NN with known weights (right)

Ali's suggestion of Fourier transform and its inverse before and after NN affects its performance.

$$\hat{\boldsymbol{y}} = [x_0 \dots x_{N-1}] \begin{bmatrix} \cos(0) & \cos(0) & \dots & \cos(0) & -\sin(0) & -\sin(0) & \dots & -\sin(0) \\ \cos(0) & \cos(\frac{2\pi}{N}) & \dots & \cos(\frac{2\pi(N-1)}{N}) & -\sin(0) & -\sin(\frac{2\pi}{N}) & \dots & -\sin(\frac{2\pi(N-1)}{N}) \\ \cos(0) & \cos(\frac{2\pi}{N}2) & \dots & \cos(\frac{2\pi(N-1)}{N}2) & -\sin(0) & -\sin(\frac{2\pi}{N}2) & \dots & -\sin(\frac{2\pi(N-1)}{N}2) \\ \cos(0) & \cos(\frac{2\pi}{N}3) & \dots & \cos(\frac{2\pi(N-1)}{N}3) & -\sin(0) & -\sin(\frac{2\pi}{N}3) & \dots & -\sin(\frac{2\pi(N-1)}{N}3) \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \cos(0) & \cos(\frac{2\pi}{N}(N-1)) & \dots & \cos(\frac{2\pi(N-1)}{N}(N-1)) & -\sin(0) & -\sin(\frac{2\pi}{N}(N-1)) & \dots & -\sin(\frac{2\pi(N-1)}{N}(N-1)) \end{bmatrix}$$

$$\hat{\boldsymbol{y}} = \boldsymbol{x} W_{Fourier}$$